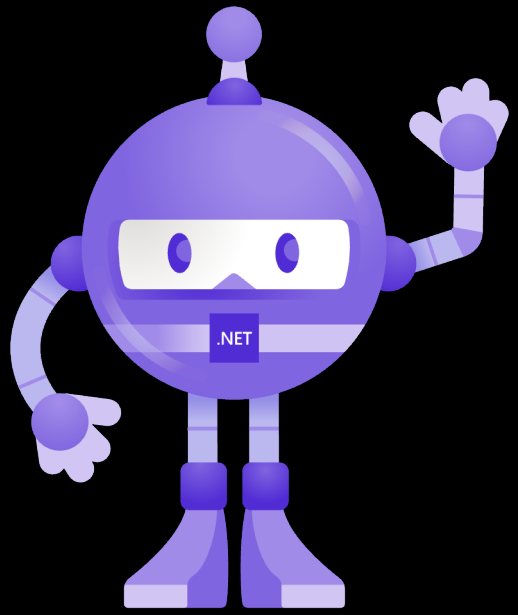
Microsoft

Build Native Apps for Any Device with .NET

Maddy Leger @maddyleger1
Program Manager, Developer Tools

David Ortinau @davidortinau
Program Manager, Developer SDKs

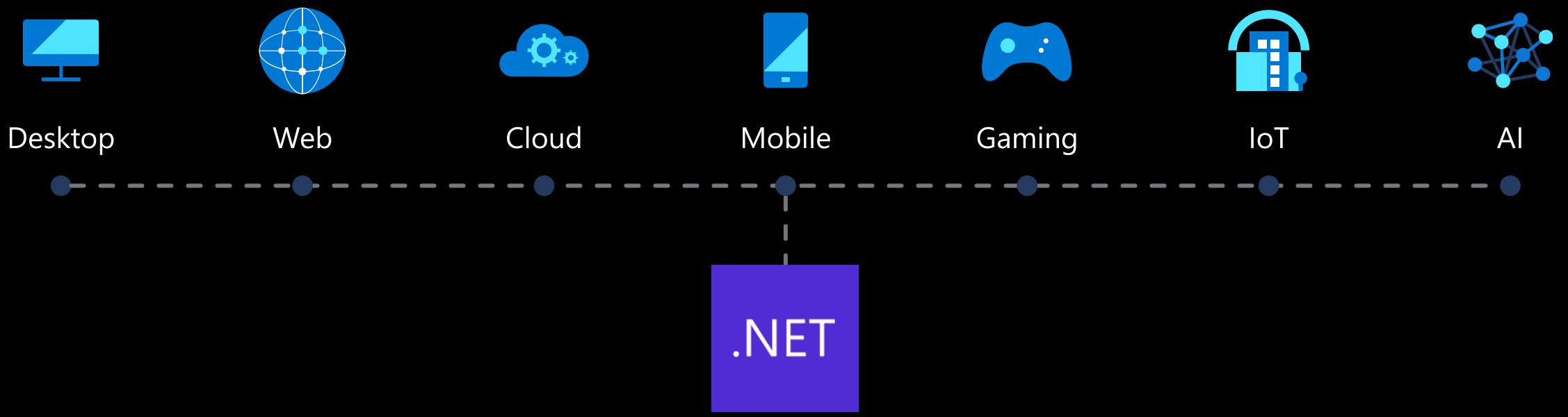


Agenda

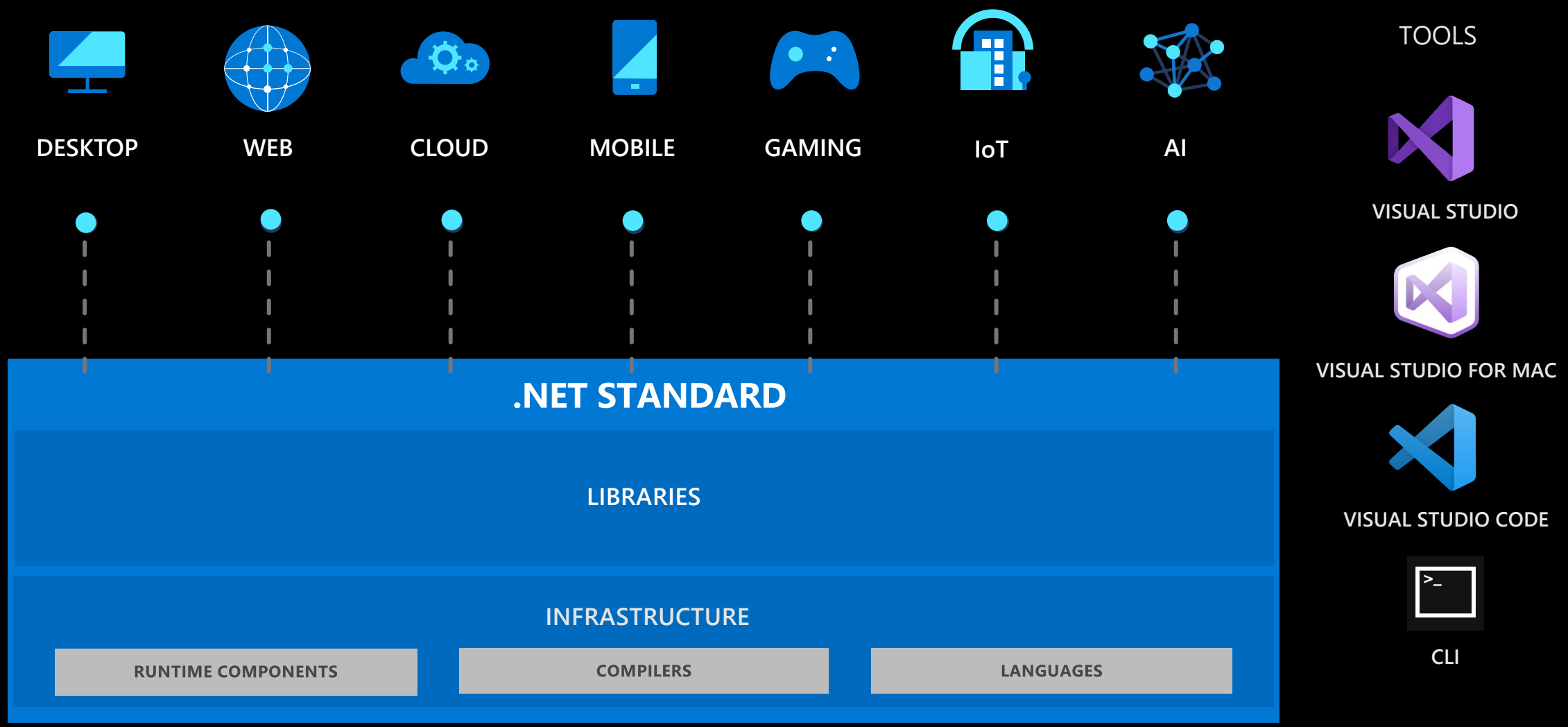
Recap

.NET

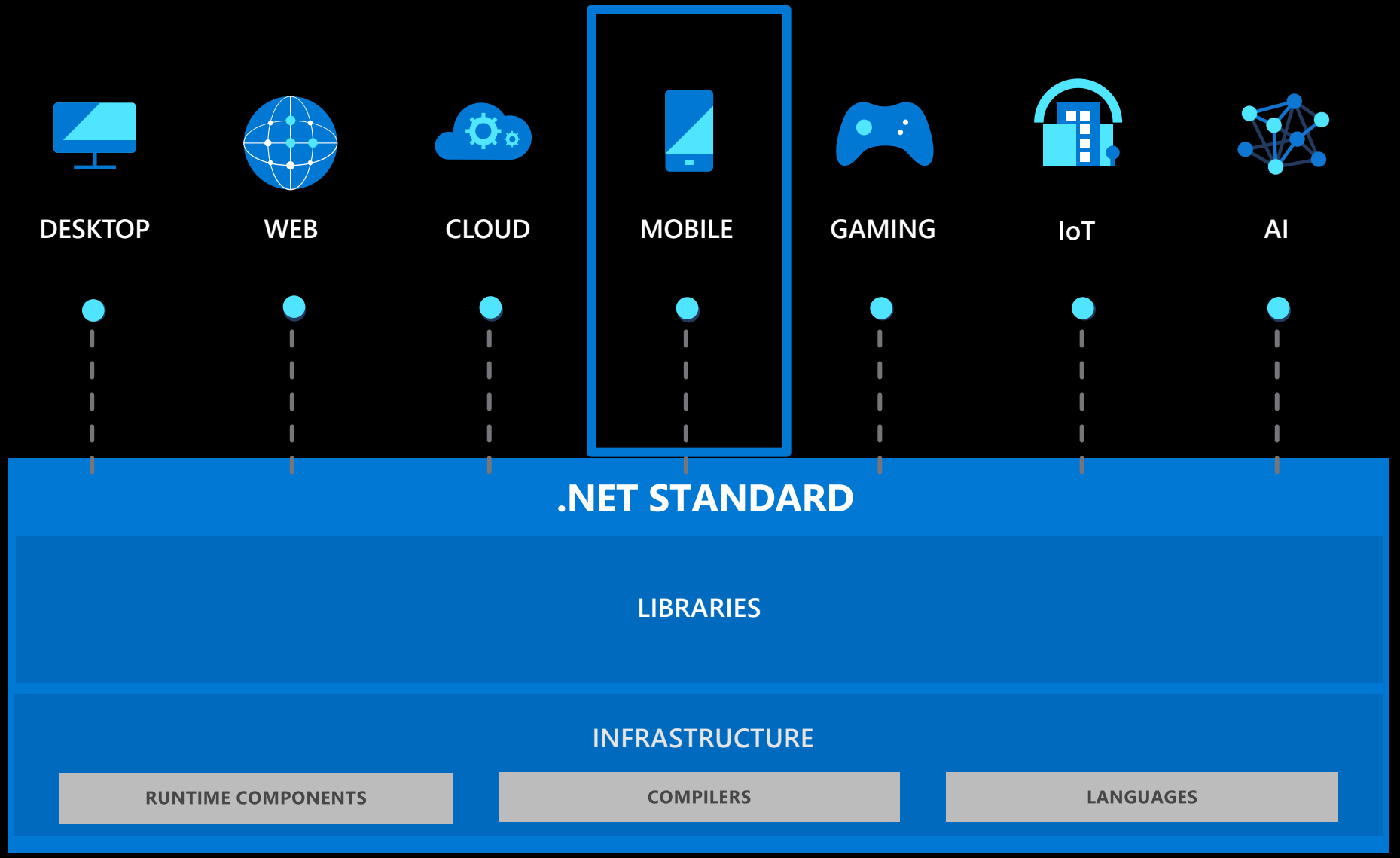
Your platform for building anything



.NET – A unified platform



.NET – A unified platform



TOOLS

VISUAL STUDIO

VISUAL STUDIO FOR MAC

VISUAL STUDIO CODE

CLI

.NET ecosystem momentum

>1.5M
.NET Core developers

in Visual Studio

#1 Most Loved
Framework

.NET Core

stackoverflow

Top 30
Highest velocity OSS projects

github.com/dotnet
github.com/aspnet

.NET foundation

Top 5
Language on GitHub

C#

7x
Faster than Node.js

ASP.NET Core

TechEmpower

40%
New to .NET are students

dot.net download survey

insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted
github.com/cncf/velocity/
octoverse.github.com/
www.techempower.com/benchmarks/#section=data-r18&hw=ph&test=plaintext

.NET adoption

Visual Studio

+1 million new monthly active .NET developers in last year

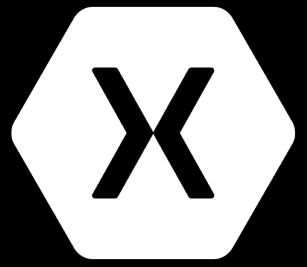
.NET Core

+600K new monthly active developers in last year

.NET on Linux

>1 million publishes from Visual Studio to Linux

An open source app platform
from Microsoft for
building modern & performant
iOS, Android, macOS,
watchOS, & tvOS apps with .NET.



Xamarin

Loved by developers, trusted by enterprise

































































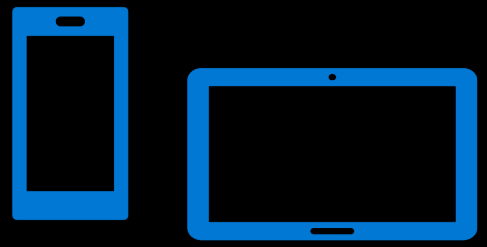








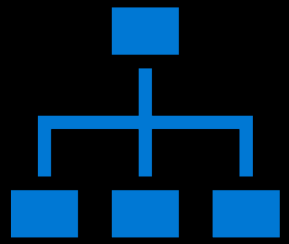
.NET mobile apps with Xamarin



iOS & Android with C#

Native performance, native APIs, native UI

Build native mobile apps with Visual Studio



Cross-Platform

Build cross-platform UI with Xamarin.Forms

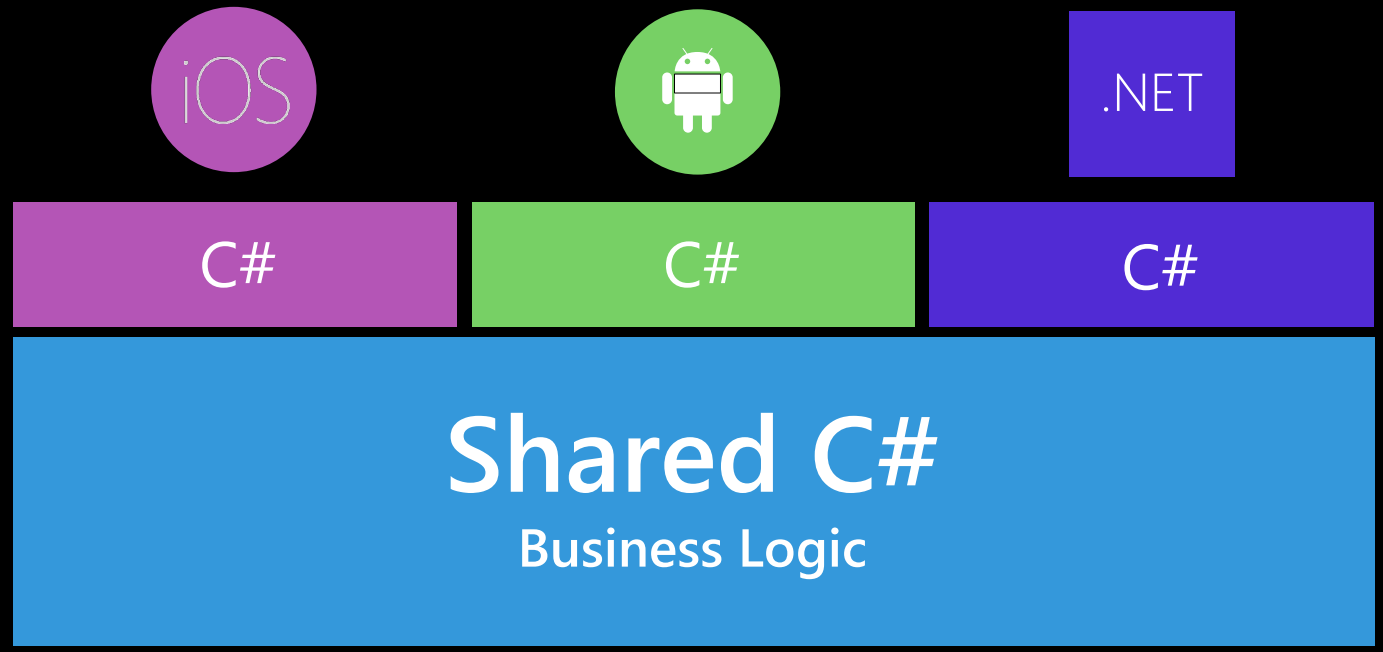
Access native APIs from shared code with Xamarin.Essentials

Share code with .NET Standard



Free & Open Source

Xamarin is part of .NET and open source on GitHub



Shared C# codebase • 100% native API access • High performance

Awesome Xamarin Component Vendors



















Developer Experience Improvements

All The Things

Fundamentals

- Smallest workload ever
- Fastest install ever
- New install location
- Lower UI delay, hang, and crash rates
- Faster build and deploy times
- Faster Android emulator
- 2x improved first build and deploy success rates

Xamarin.iOS Tooling

- Xcode 11 + iOS 13 tooling
- Faster Remote iOS Simulator
- iOS Designer
  - Custom control support
  - Dynamic library support (SkiaSharp)
  - Xcode design preference
- Xamarin Hot Restart

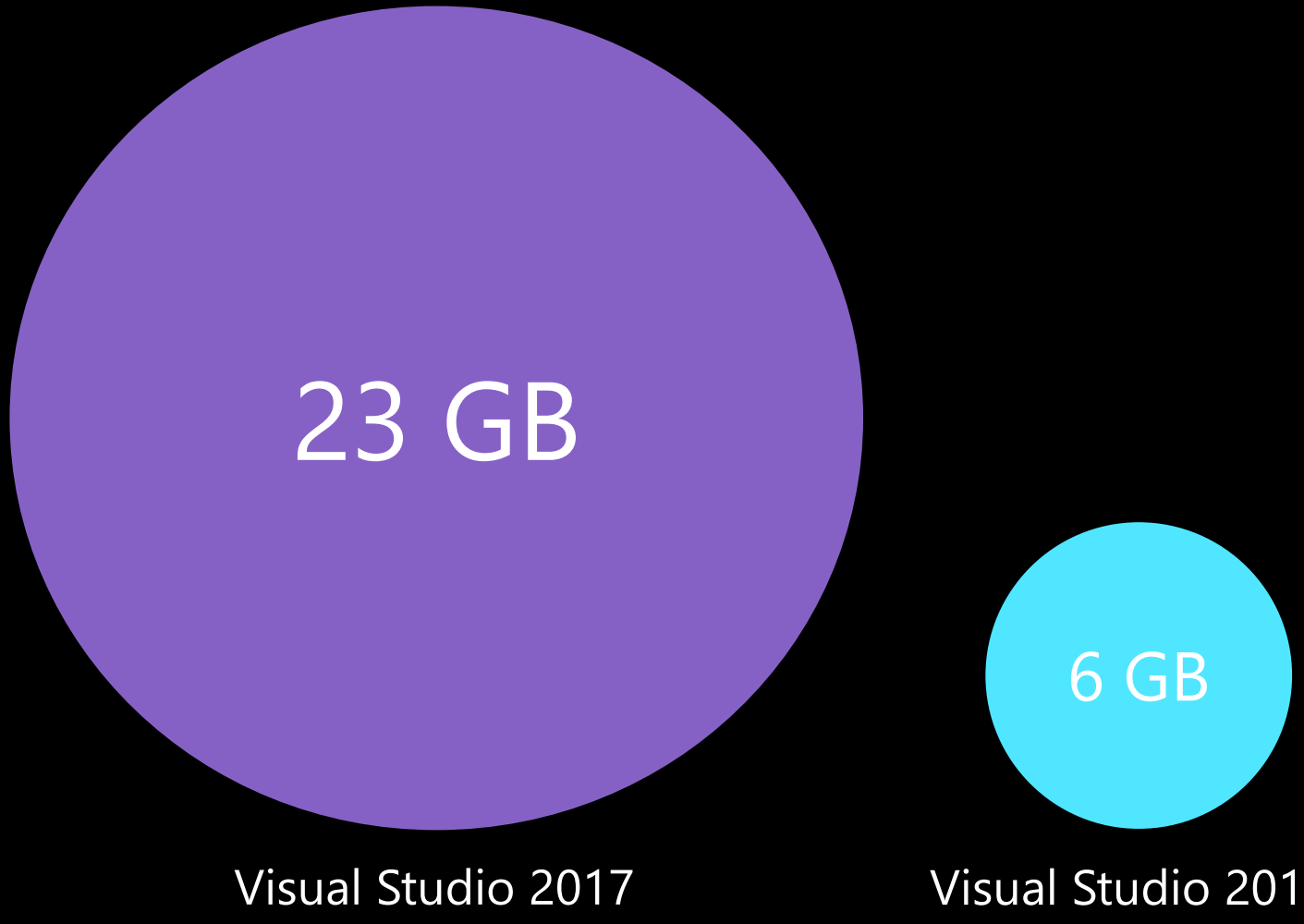
Xamarin.Android Tooling

- Automatic Android SDK management + repair
- AXML Editing
  - New editor experience on Windows and macOS
  - Snippets
  - IntelliSense for all resource file types
  - Code navigation
  - Linting capabilities
- Android Designer
  - Quick editing experience
  - Split view
  - Revamped property panel
  - Use .xml extension
- Android X migration experience
- Profiled AOT + more Android optimizations
- Android App Bundle support
- Android Q

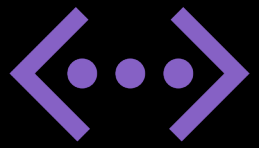
Xamarin.Forms Tooling

- XAML Editing
  - IntelliCode
  - Property Panel
  - New editor in Visual Studio for Mac
- XAML Hot Reload for Xamarin.Forms
  - Multiple debug targets
- XAML Previewer
  - Design time data
  - Real device sizes

Xamarin is now over 2x faster to install



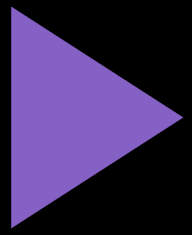
Boosting Developer Productivity



Code faster with IntelliCode and improved Intellisense in rebuilt C# and XAML editors (and soon Android XML).

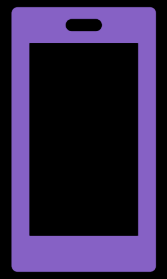
Code further for multiple platforms with multi-targeting.

Code deeper into source code via SourceLink.



Build & Deploy faster with optimized compilers.

- First: -21.95%
- Incremental: -29.66%
- Deploy: -50.83%
- Aapt2 & Incremental APK Builds



Startup faster with Startup Tracing; code paths are optimized with Ahead-of-Time compilation to save 60%

Reduce app size using App Bundles.



Iterate faster with XAML Hot Reload for instant updating of view changes, and Hot Restart for optimized IDE to Device changes to C#, XAML, and resources all without recompilation.

Demo: Be more productive right now! 🔥





Trevor Tirrell
@TrevorTirrell

Replying to @maddyleger1 and @JamesMontemagno

Enjoying🔥🔄 Nicely done. Thanks!!

1:52 PM · Sep 27, 2019 · Twitter Web App



Shaboo @beckhamsh · Oct 13, 2019
The new **#HotReload** feature for **#xamarin** is amazing
Thanks @xamarinhq @VisualStudio

1



Shaun Hevey @shaunhevey · Aug 2, 2019
Trying out **#xamarin** XAML **#HotReload** for Xamarin.Forms preview on Visual Studio for Mac. My early thoughts are, this is fantastic!

6



Umit AYDIN
@umitnuri

Replying to @pierceboggan

This is one of the biggest contribution to my development cycle welldone team 👍👏👏

12:46 PM · Dec 4, 2019 from Melbourne, Victoria · Twitter for iPhone



Geir "Golf80" Jónsson @GeirJonsson · Jan 29
Are you kidding me with that **#hotreload** in **#Xamarin** forms - amazing improvement #microsoft :) #devlife #golf80

1 2 9



Norman Mackay @normanmackay3 · Jan 23
Replying to @dylbot9000 @rlittlesii and 2 others
Getting HotReload working was awesome...I paid for a LiveXAML licence long before. Cool to see it coming to the tooling. Saves a tonne of time.

3



Carlos Campos @codercampos · Jan 23
After reviewing in private this feature, now is finally live! **#HotRestart** is one of the most anticipated feature any Xamarin developer is waiting and now you can grab it in the preview channel!

#Xamarin #MobileDev #VisualStudio



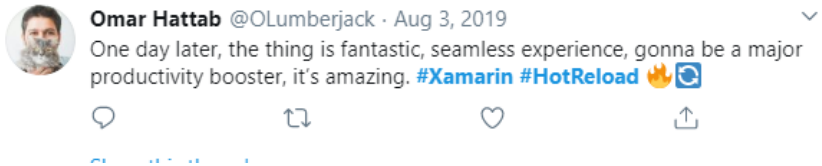
Dylan Berry 🥒🚇
@dylbot9000

Replying to @rlittlesii @maddyleger1 and @XamDevSummit

It really has been improving significantly. Hot reload, CollectionView, CheckBox, the list goes on. It shows that the team is listening to developers and making changes to improve their experience and make them more productive.

12:56 PM · Jan 23, 2020 · Twitter Web App

Let's say it out loud, it is working great in Xamarin.Forms. It's really boosting development.



Omar Hattab @OLumberjack · Aug 3, 2019
One day later, the thing is fantastic, seamless experience, gonna be a major productivity booster, it's amazing. **#Xamarin #HotReload** 🔥🔄



Brett Rigby @brettrigby · Jan 30
Am loving the XAML Hot Reload in Xamarin.Forms! An excellent time-saver! Have been using it for a few days now on an Android app, and can't think how much it's helped me already.

#xamarin #xamarinForms #xaml **#hotreload** #xamlHotReload #xamdev @pierceboggan @msdevUK #Android

1 10



Chris Roberts @chrismrobertsuk · Aug 2, 2019
Well @maddyleger1 @davidortinau @JamesMontemagno, **#Xamarin #HotReload** is awesome! Thank you for my preview invitation. Love love love it, thanks to the whole team for your hard work 😎 🔥🔄

But yes the deployment is so much faster. This changes everything for me



.Morten
@dotMorten

Did you sign up for Xamarin Hot Restart preview yet? Been testing it for a while & LOVING being able to deploy directly to an iOS device without a Mac. No more xcode out of date, or needing a mac or dealing with connection issue. Dev loop also WAY faster.



Clifford Agius
@CliffordAgius

Replying to @pierceboggan

Been taking Hot Restart for a spin today again very impressed. I turned my Macbook that lives on the shelf off just to be sure you wasn't cheating. 😂

12:59 PM · Dec 4, 2019 · Twitter for Android

" Hot reload has made me so much happier (and more productive), I couldn't ask for more."



Dan Siegel @DanJSiegel · Nov 21
Been using #Xamarin **Hot Restart** for about a month now... it is absolutely amazing... even though I primarily work on my Mac with Windows 10 running in Parallels, it's still utterly amazing that I can deploy to my iPhone from VS on Windows with no Mac Connection!!!

3 1 15

Hot Restart

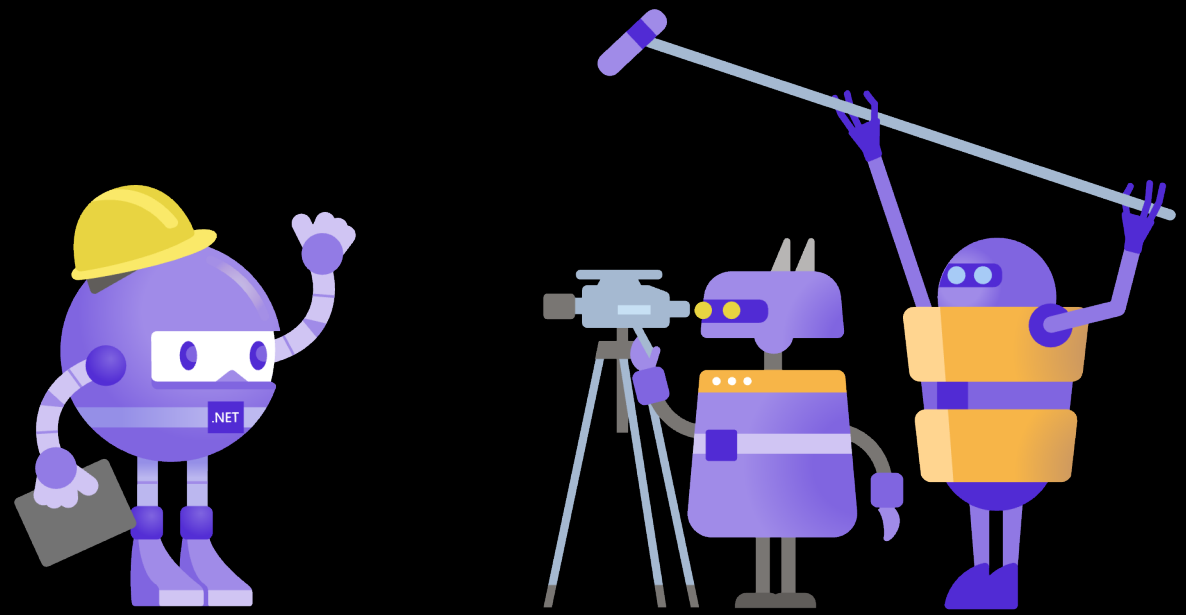
Test changes made to your app, including multi-file code edits, resources, and references while using a much faster build and deploy cycle.

With Hot Restart, debug your iOS app built with Xamarin.Forms on a device connected to your Windows machine allowing for a much faster inner development loop.

	Initial Build	Initial Deploy	Incremental Build	Incremental Deploy
Hot Restart	22.8s	39.5s	2.5s	7.3s
w/o Hot Restart	172.7s	26.2s	37.2s	14.4s

- Now:
  - iOS on Windows
- Next:
  - Android on Mac/Windows
  - iOS on Mac
  - Integrated F5 experience

SDK Improvements



In Step and Up to Date



Xcode 11.4 support
out now



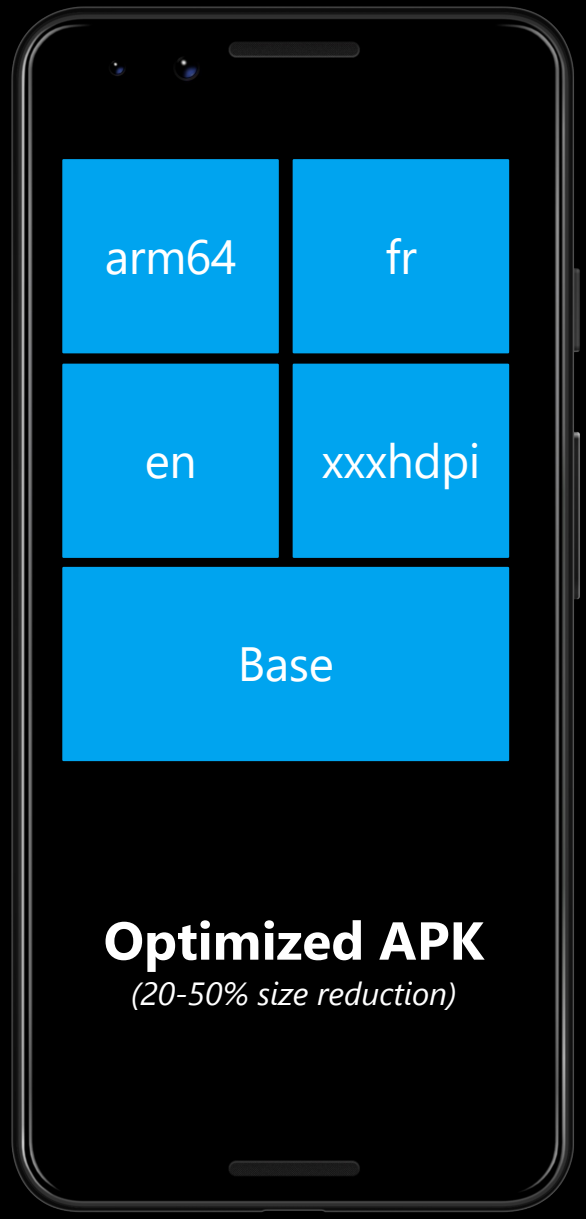
Android 11 support
available in 16.6

Android App Bundles

- Specific Languages
- Proper Resource Screen Densities
- Best Match ABIs







AndroidX



AndroidX





Android 11 – VS 2019 16.6

What's New in Xamarin.Forms

Presented by You



"I am adding 5-7 new good looking ui samples in these weeks to the list. Isn't it amazing?!"

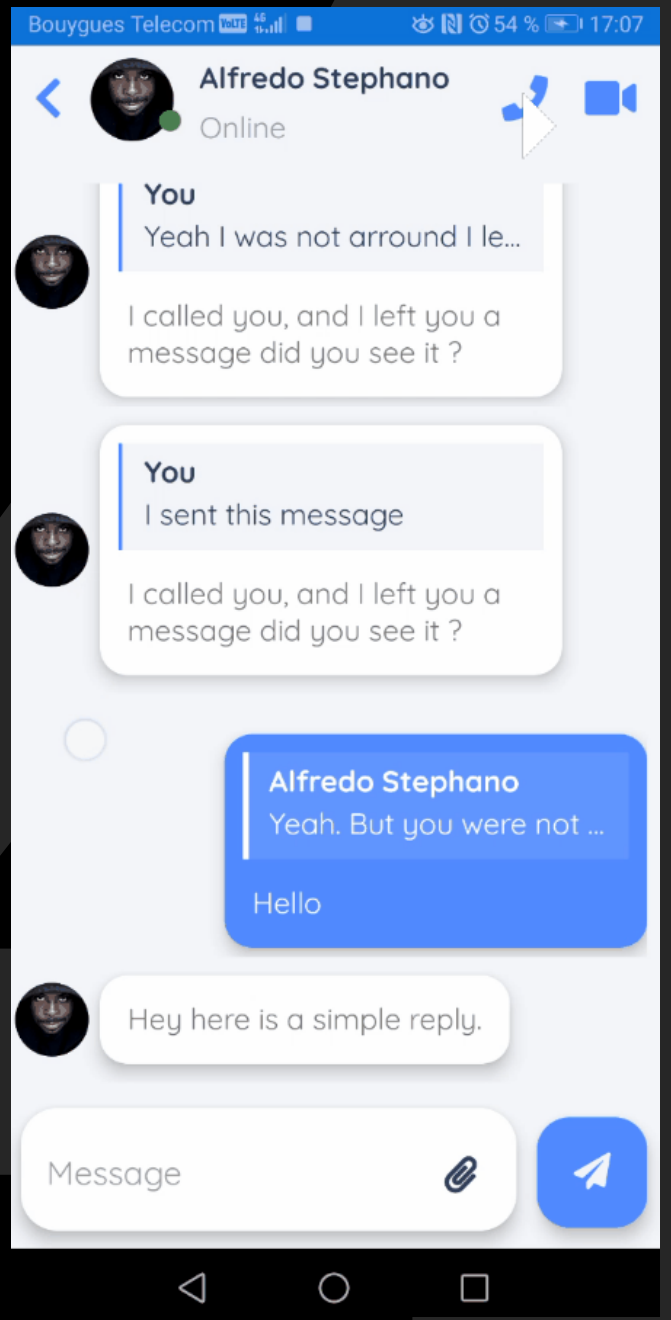
aka.ms/xf-good-ui



Javier Suárez

CollectionView

```
<CollectionView ItemsSource="{Binding Messages}"
                IsGrouped="True"
                BackgroundColor="{StaticResource PrimaryColor}"
                x:Name="MessagesCollectionView">
  <CollectionView.GroupHeaderTemplate>
    <DataTemplate>
      <v:MessageHeaderView />
    </DataTemplate>
  </CollectionView.GroupHeaderTemplate>
  <CollectionView.ItemTemplate>
    <dataTemplateSelector:ChatBubbleTemplateSelctor />
  </CollectionView.ItemTemplate>
</CollectionView>
```

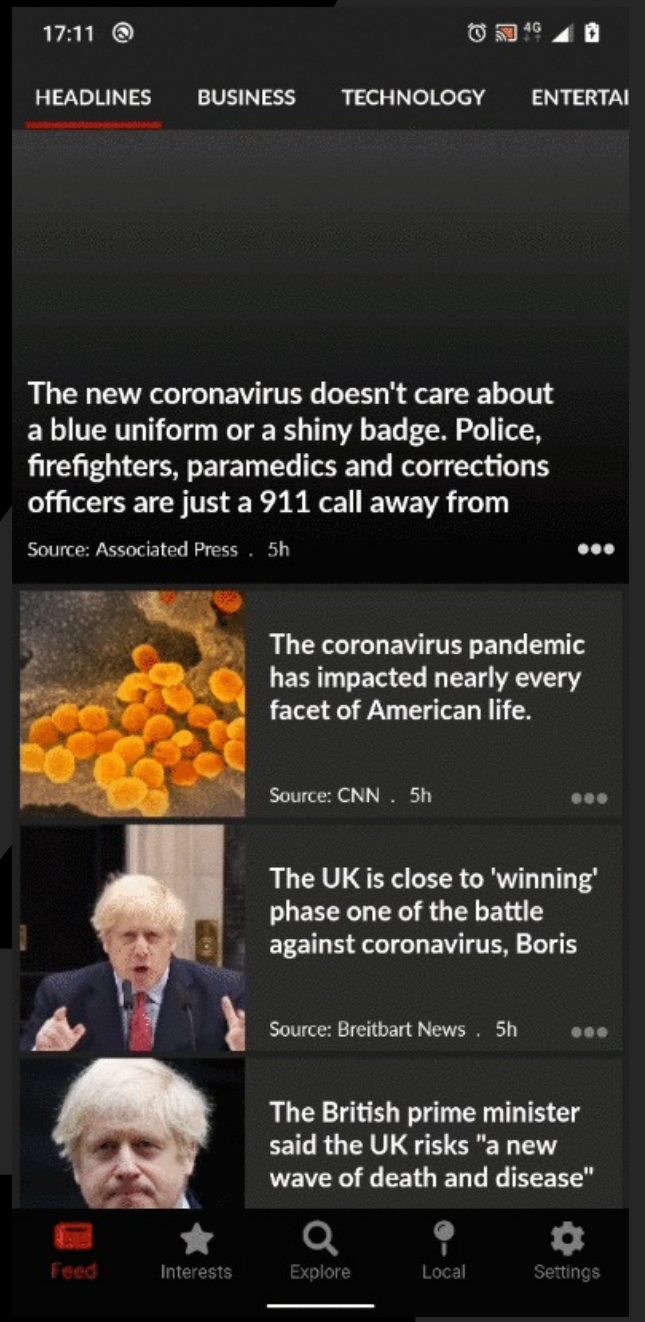


https://github.com/DamienDoumer

Shell

```
<TabBar>
  <Tab Title="Feed"
    Icon="{StaticResource NewspaperIcon}"
    Route="mainFeed">
    <ShellContent
      ContentTemplate="{DataTemplate views:MainFeed}" />
  </Tab>
  <Tab Title="Interests"
    Icon="{StaticResource StarIcon}"
    Route="interests">
    <ShellContent
      ContentTemplate="{DataTemplate views:MyInterests}" />
  </Tab>
  <Tab Title="Explore"
    Icon="{StaticResource SearchIcon}"
    Route="explore">
    <ShellContent
      ContentTemplate="{DataTemplate views:Explore}" />
  </Tab>
```

https://github.com/Elisha-Misoi

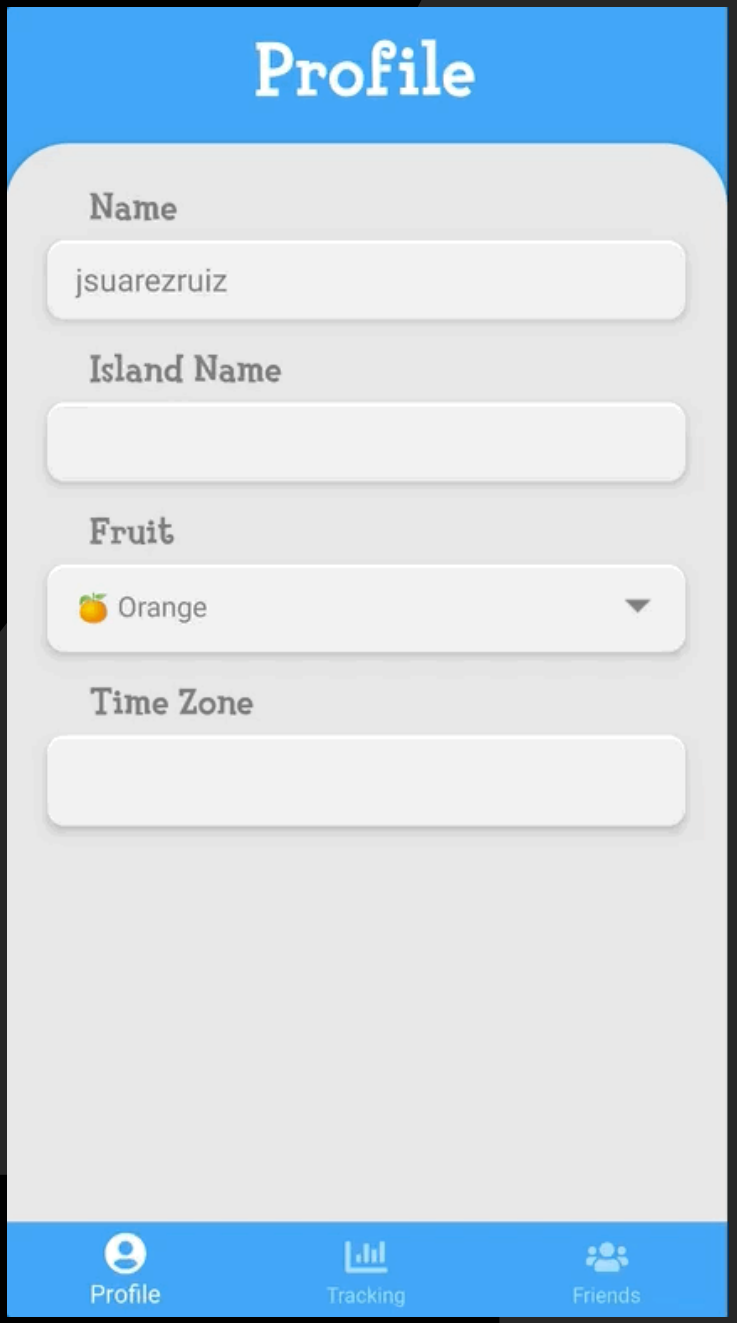


Visual

```
<CheckBox
   Visual="Material"
   IsChecked="{Binding SelectedDay.FirstPurchase}"
   Color="{StaticResource NavigationPrimary}" />

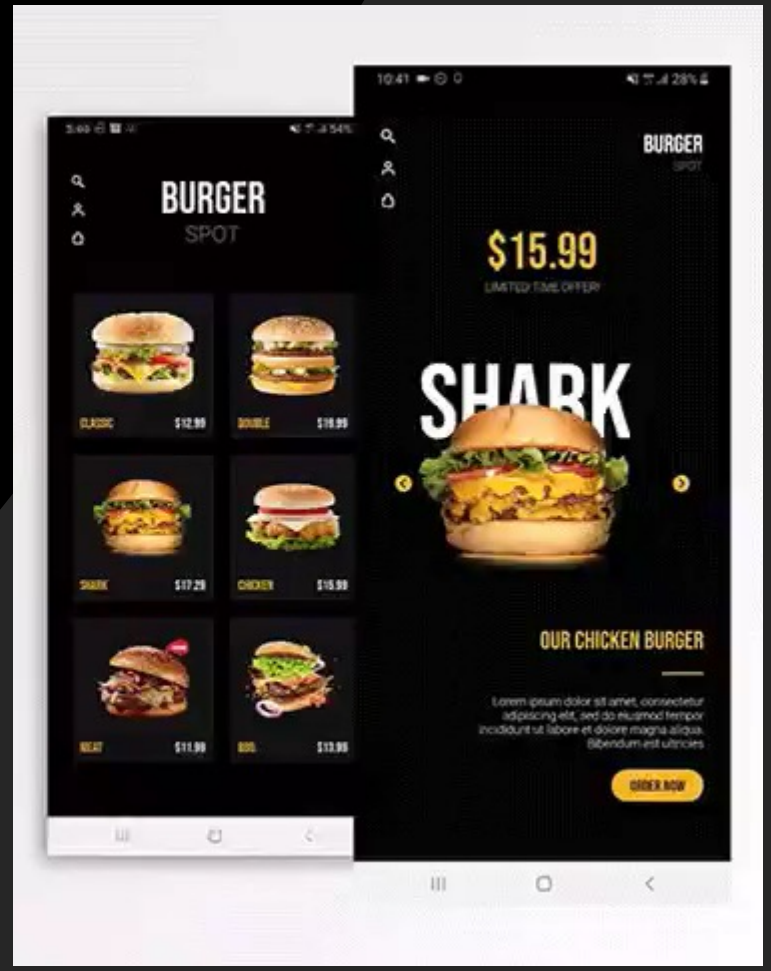
<Style x:Key="ButtonRoundedCorners" TargetType="Button">
   <Setter Property="Visual" Value="Material"/>
   <Setter Property="BorderWidth" Value="2"/>
   <Setter Property="CornerRadius" Value="22"/>
   <Setter Property="HeightRequest" Value="44"/>
   <Setter Property="Padding" Value="12,0"/>
</Style>
```

https://github.com/jamesmontemagno



CarouselView

```
<CarouselView
    ItemsSource="{Binding Burgers}"
    HeightRequest="260"
    HorizontalOptions="FillAndExpand"
    CurrentItem="{Binding SelectedBurger}"
    IsScrollAnimated="True"
    IsSwipeEnabled="True">
    <CarouselView.ItemTemplate>
        <DataTemplate>
            <t:BurgerItem />
        </DataTemplate>
    </CarouselView.ItemTemplate>
</CarouselView>
```



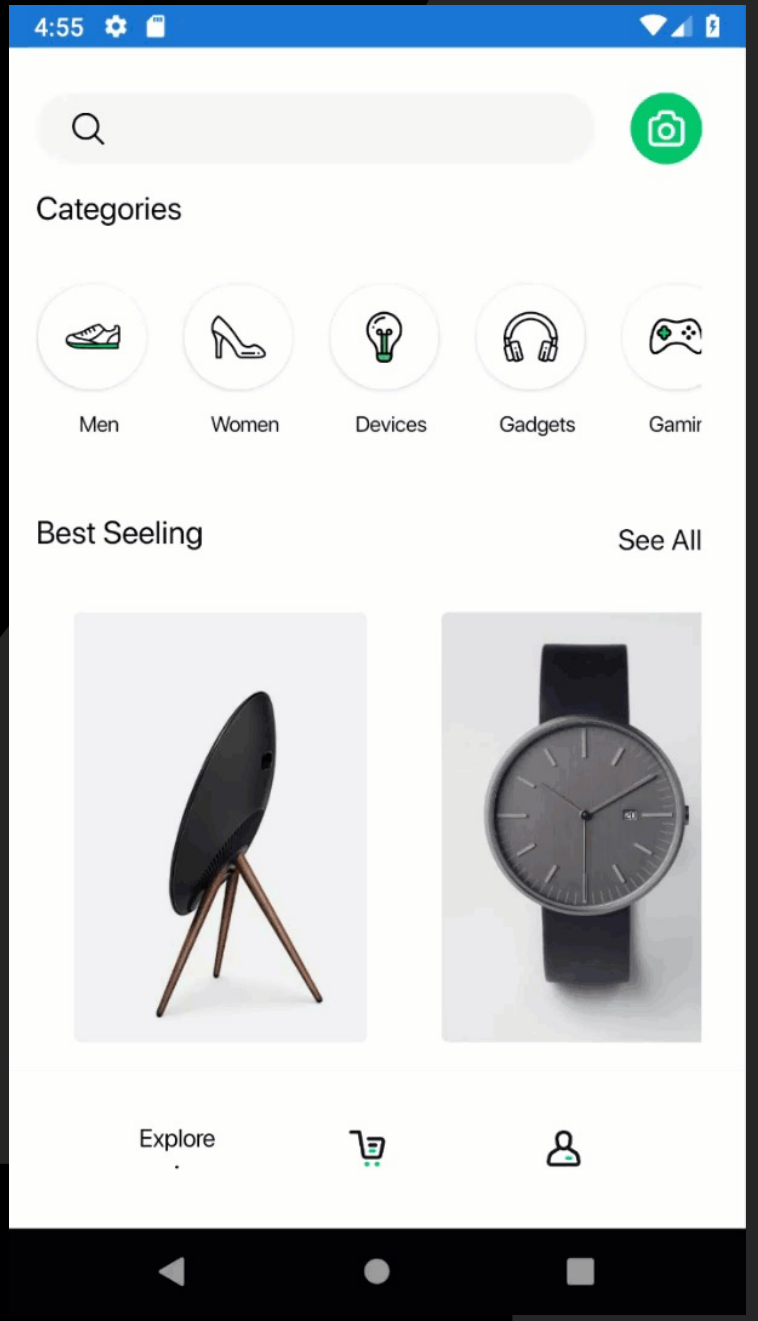
https://github.com/devcrux

SwipeView

```
<SwipeView>
  <SwipeView.LeftItems>
    <SwipeItems>
      <SwipeItem
        BackgroundColor="#FFC107"
        Invoked="OnFavoriteSwipeItemInvoked"
        Text="Favorite" />
    </SwipeItems>
  </SwipeView.LeftItems>
  <SwipeView.RightItems>
    <SwipeItems>
      <SwipeItem
        BackgroundColor="#FF3D00"
        Command="{Binding DeleteCommand}"
        CommandParameter="{Binding}"
        Text="Delete" />
    </SwipeItems>
  </SwipeView.RightItems>

  <!-- Content -->

</SwipeView>
```



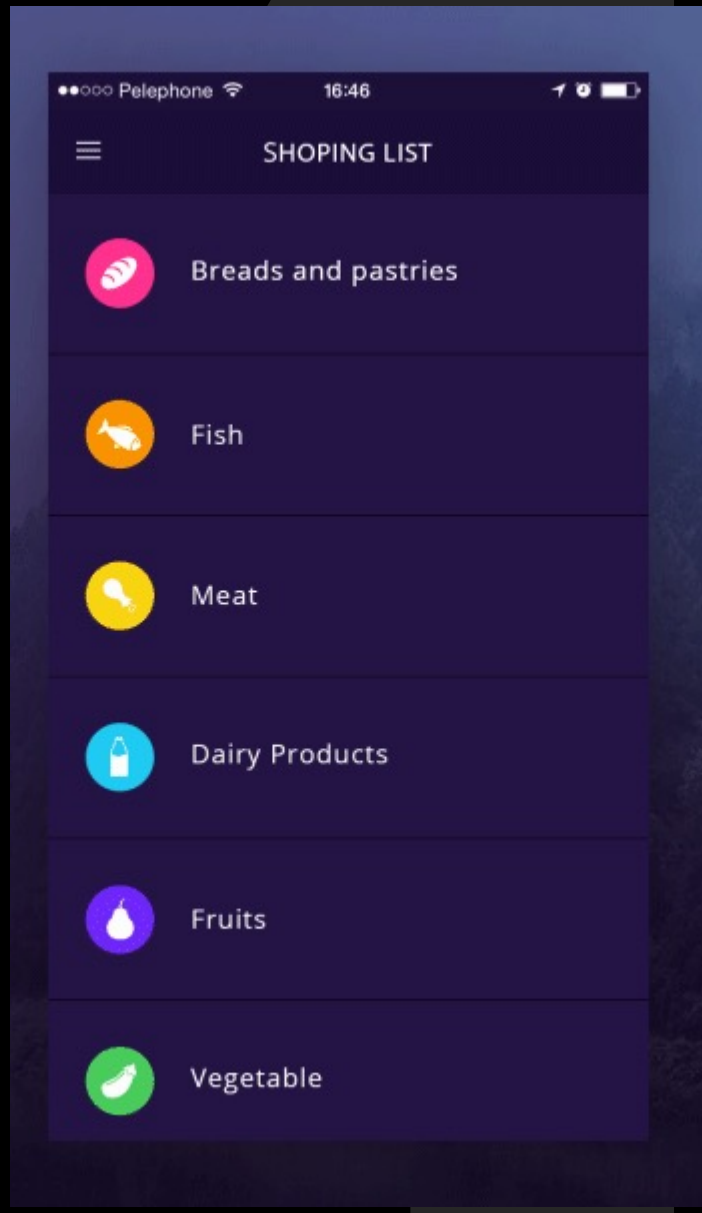
https://github.com/exendahal

Expander

```
<Expander
   ExpandAnimationEasing="{x:Static Easing.Linear}"
   CollapseAnimationEasing="{x:Static Easing.Linear}"
   IsExpanded="{Binding IsDetailVisible, Mode=TwoWay}">
   <Expander.Header>
      <v:HeaderView />
   </Expander.Header>

   <!-- content -->

</Expander>
```

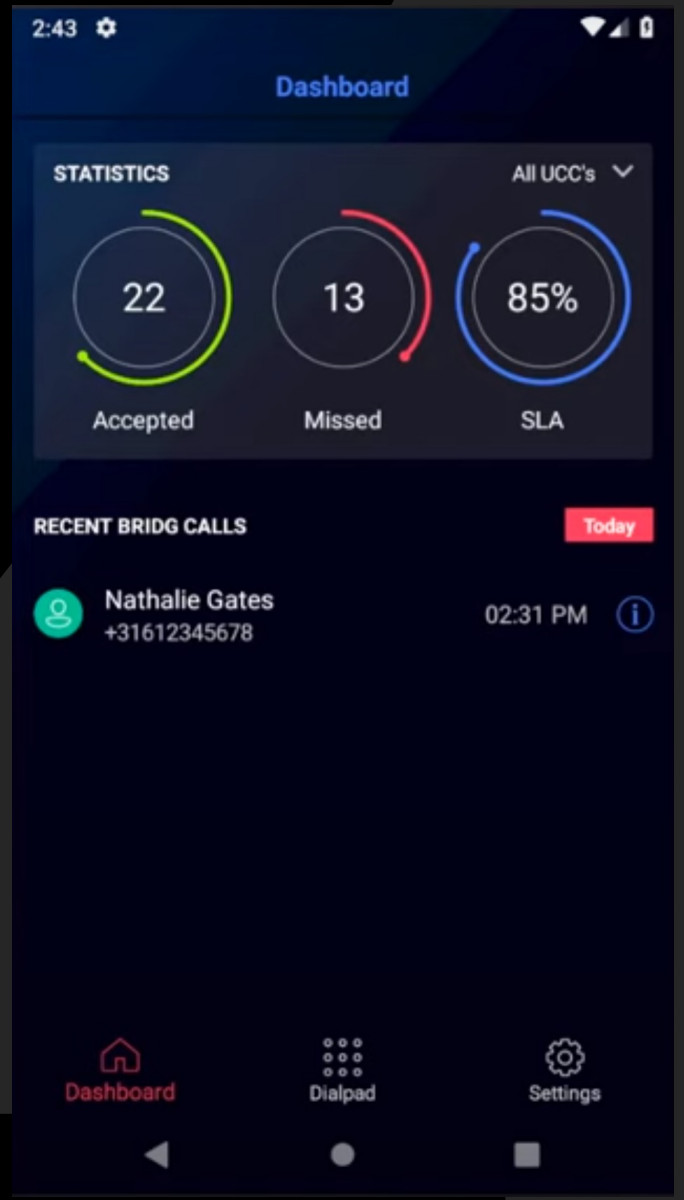


https://github.com/jsuarezruiz

C# UI

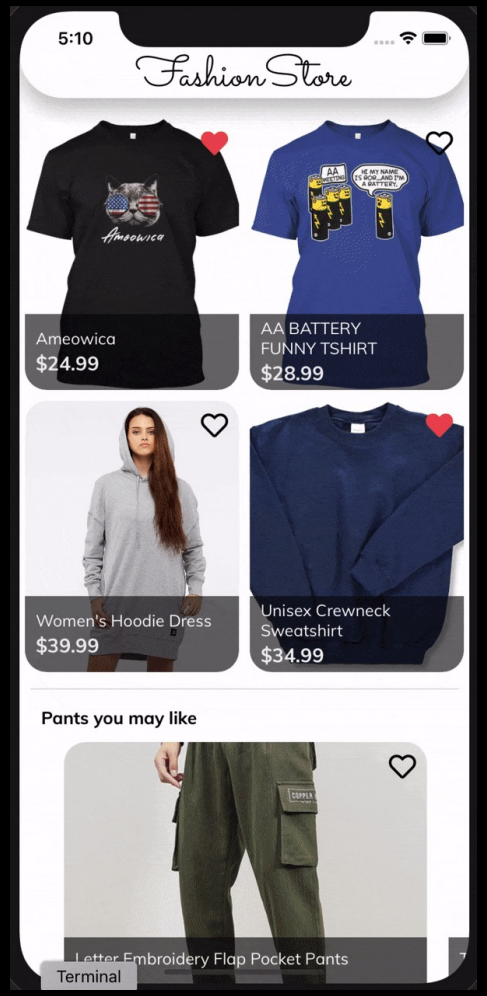
```
Content = new Grid {
    RowSpacing = 0,
    RowDefinitions = Rows.Define(
        (PageRow.Header, Auto),
        (PageRow.Body   , Star)
    ),
    Children = {
        new ScrollView { Content = new StackLayout { Children = {
            new Button { Text = nameof(RegistrationCodePage) }
                .Style (FilledButton)
                .FillExpandH ()
                .Margin (PageMarginSize)
                .Bind (nameof(vm.ContinueToRegistrationCommand)),

            new Button { Text = nameof(NestedListPage) }
                .Style (FilledButton)
                .FillExpandH ()
                .Margin (PageMarginSize)
                .Bind (nameof(vm.ContinueToNestedListCommand)),
```

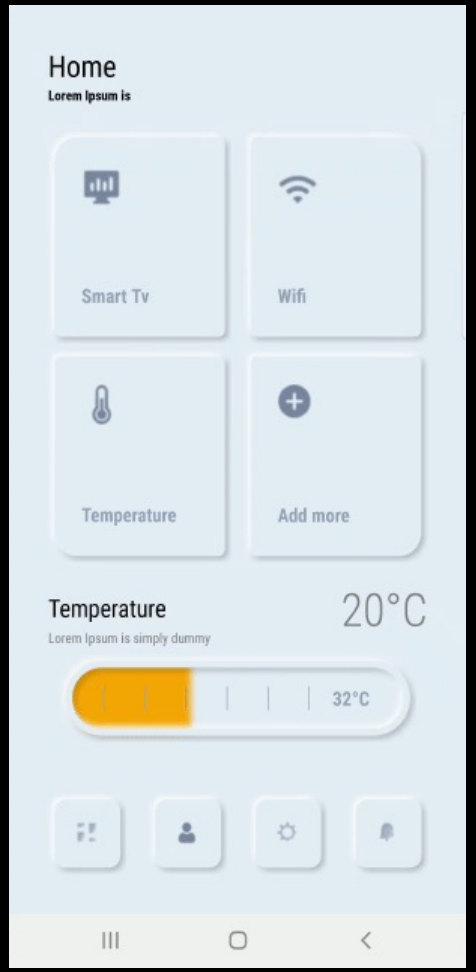


https://github.com/VincentH-Net

Plugins



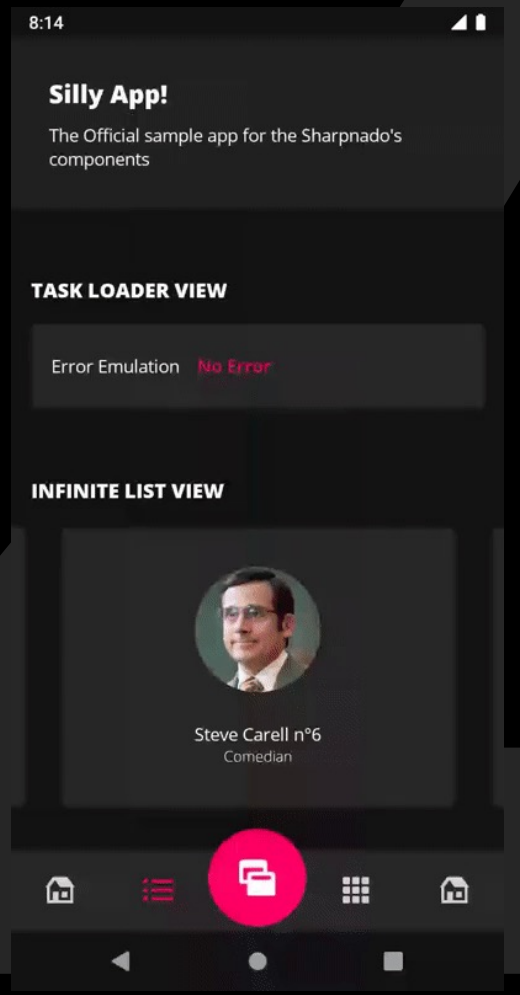
SharedTransitions
github.com/valentineg8



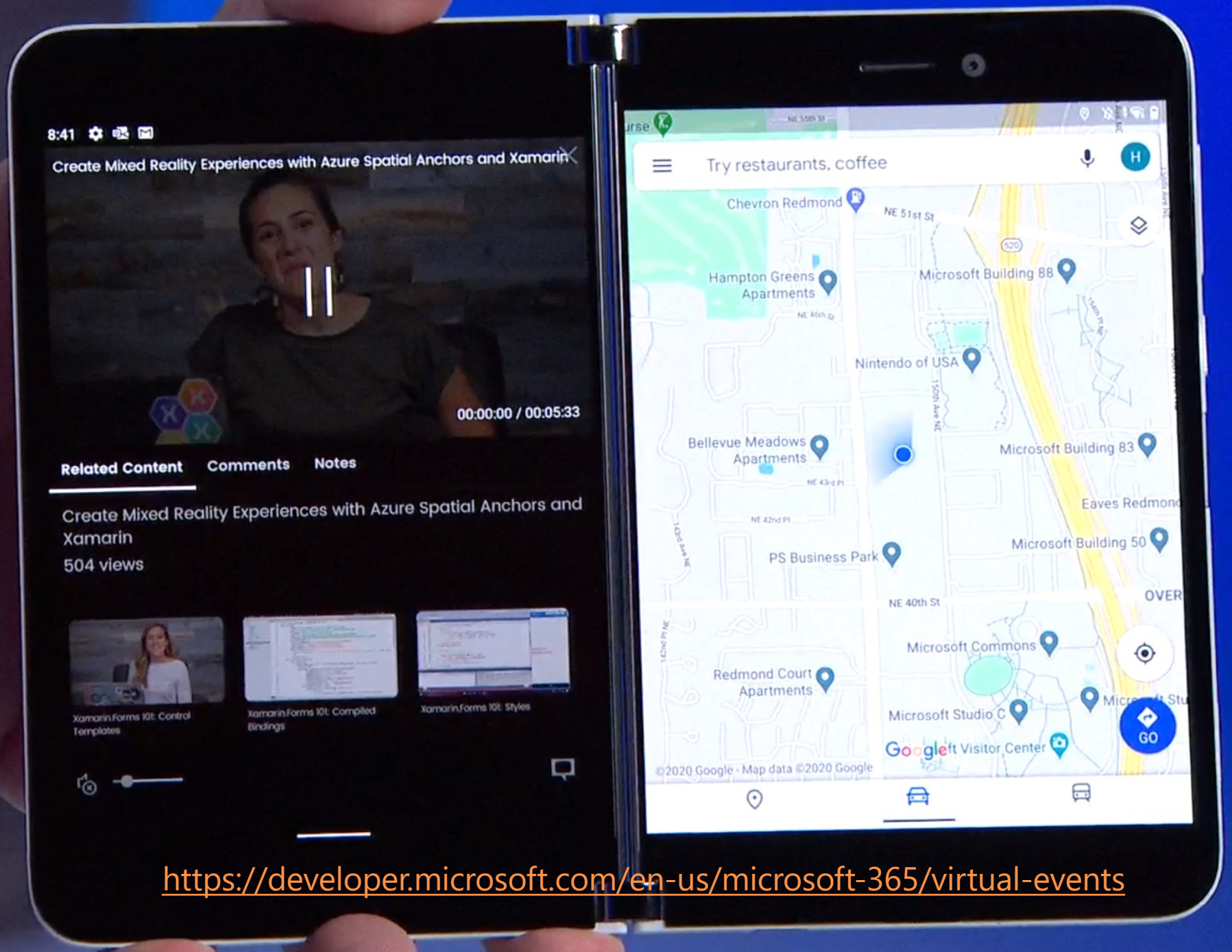
NeoControls
github.com/felipebaltazar



PancakeView
github.com/Altevir

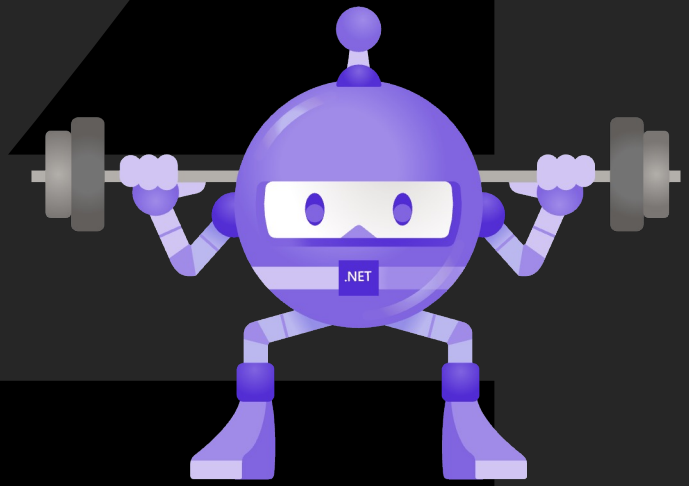


Sharpnado
github.com/roubachof



https://developer.microsoft.com/en-us/microsoft-365/virtual-events

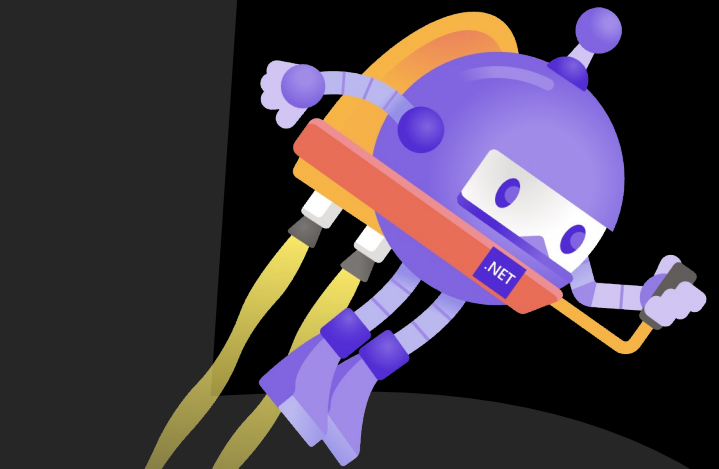
Demo: Dual Screen



Last Year in Xamarin.Forms

- Accessibility
- BindableLayout EmptyView
- C# UI Extensions
- Character Spacing
- CheckBox
- EmbeddedFonts
- FontImageSource
- GIF Animations
- HTML Label
- IndicatorView
- Maps+
- MediaElement
- Modal Prompts
- Multi-Binding
- RadioButton
- RefreshView
- RelativeSource Binding
- Shell Modals
- SourceLink
- VisualStateManager Target
- WKWebView
- + Much More!

Coming in Xamarin.Forms 5



Shell 2.0

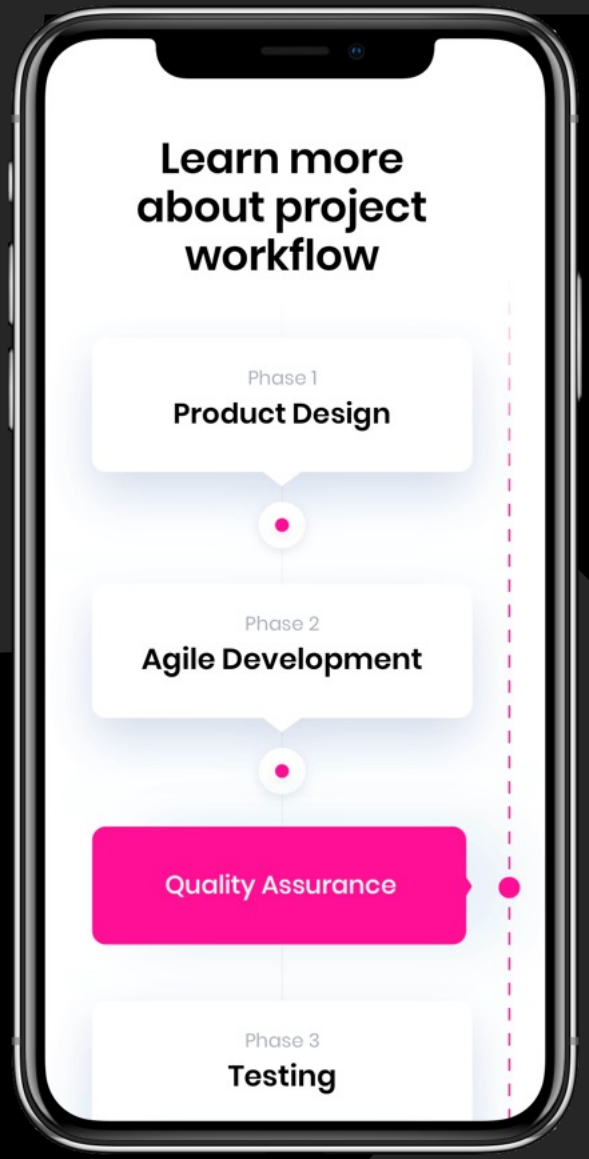
```
<Shell>
    <Shell.Routes>
        <Route Name="onboarding" ContentTemplate="{DataTemplate OnboardingView}"/>
        <Route Name="login" ContentTemplate="{DataTemplate LoginView}"/>
        <Route Name="forgot" ContentTemplate="{DataTemplate ForgotPasswordView}"/>
        <Route Name="support">
            <TabBar>
                <Tab Title="Open Issues">
                    <ShellContent ContentTemplate="{DataTemplate IssuesView}"/>
                </Tab>
                <Tab Title="Resolved Issues">
                    <ShellContent ContentTemplate="{DataTemplate ResolvedIssuesView}"/>
                </Tab>
            </TabBar>
        </Route>
    </Shell.Routes>

    <TabBar>
        <Tab
            Title="Home"
            Icon="{StaticResource HomeIcon}">
            <ShellContent ContentTemplate="{DataTemplate HomeView}"/>
        </Tab>
        <Tab
            Title="Feed"
            Icon="{StaticResource FeedIcon}">
```

Shapes & Paths

```
<!-- Lines and Dots -->
<Line X1="0" X2 ="300" Stroke="Pink"
   StrokeThickness="2" StrokeDashArray="1,1"/>
<Ellipse HeightRequest="12" WidthRequest="360" Stroke="Pink" />

<!-- Conversation Bubble -->
<Grid Margin="12">
 <BoxView
   Grid.RowSpan="2"
   Fill="{StaticResource AccentBrush}"/>
 <Label Text="{Binding Text}">
 <Label
   Grid.Row="1"
   Text="{Binding =Timestamp}"/>
 <Path Grid.Row="2"
   Data="m 0,0 l 16,0 l 0,16 l -16,-16"
   Fill="{StaticResource AccentBrush}"
   Margin="0,0,5,0" />
</Grid>
```

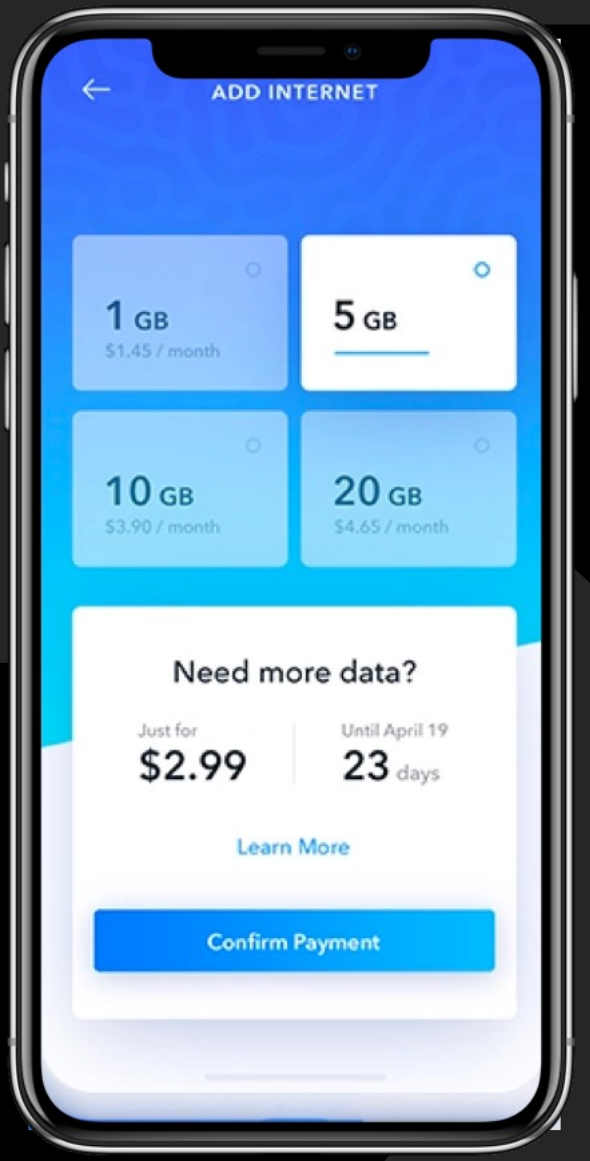


Gradient Brush

```
<LinearGradientBrush
        x:Key="HorizontalLinearGradient"
        StartPoint="0, 0"
        EndPoint="1, 0">
        <LinearGradientBrush.GradientStops>
                <GradientStop Color="#1C7EFF" Offset="0.1" />
                <GradientStop Color="#3771F7" Offset="0.6" />
                <GradientStop Color="#3877F6" Offset="1.0" />
        </LinearGradientBrush.GradientStops>
</LinearGradientBrush>

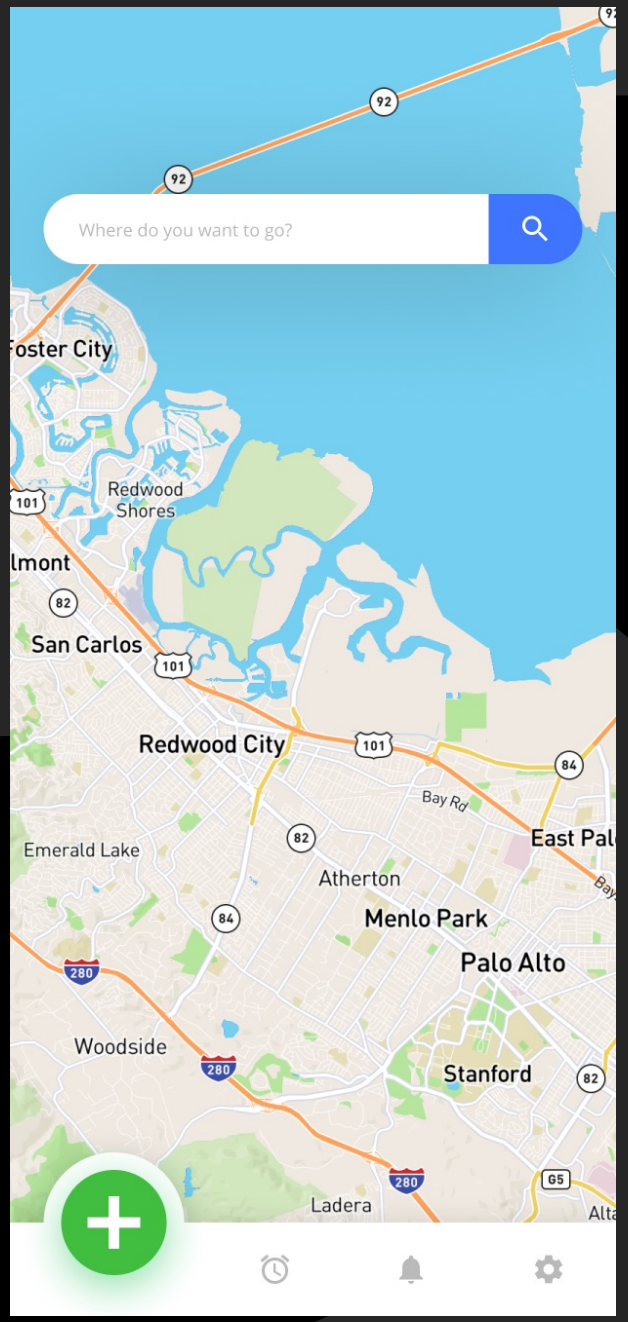

<NavigationPage
        BarBackground="{StaticResource HorizontalLinearGradient}">

</NavigationPage>
```



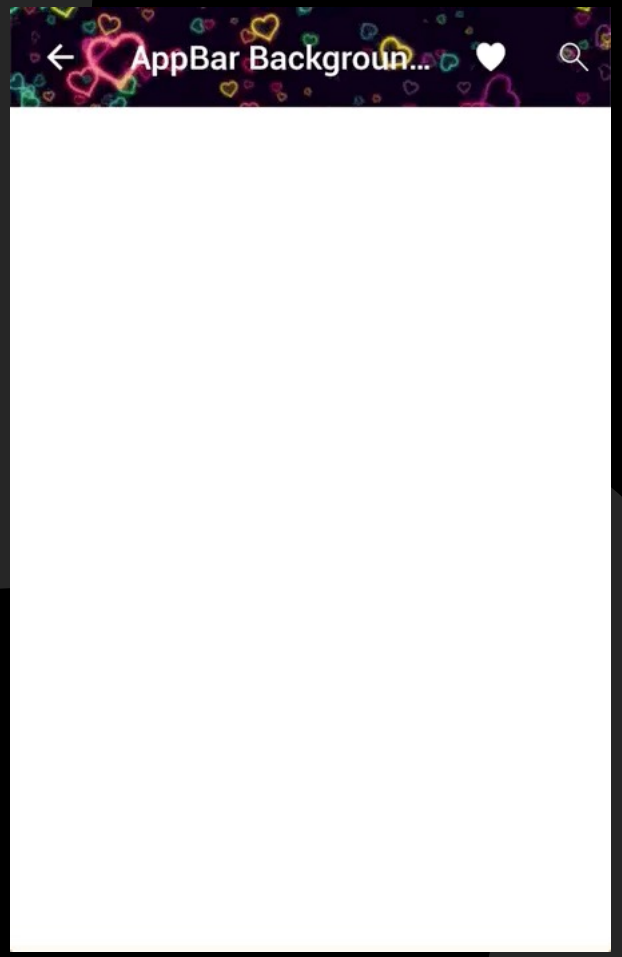
TabView

```
<TabView
        Style="{StaticResource CustomTabStyle}">
        <TabViewItem
                ControlTemplate="{StaticResource FabTabItemTemplate}"
                Style="{StaticResource TabItemStyle}"
                Icon="{StaticResource PlusIcon}"
                Command="{Binding AddCommand}"/>
        <TabViewItem
                ControlTemplate="{StaticResource TabItemTemplate}"
                Style="{StaticResource TabItemStyle}"
                Icon="{StaticResource ClockIcon}"
                ContentTemplate="{DataTemplate ClockView}"/>
        <TabViewItem
                ControlTemplate="{StaticResource TabItemTemplate}"
                Style="{StaticResource TabItemStyle}"
                Icon="{StaticResource AlertIcon}"
                ContentTemplate="{DataTemplate AlertView}"/>
        <TabViewItem
                ControlTemplate="{StaticResource TabItemTemplate}"
                Style="{StaticResource TabItemStyle}"
                Icon="{StaticResource SettingsIcon}"
                ContentTemplate="{DataTemplate SettingsView}"/>
</TabView>
```



AppBar

```
<ContentPage>
  <AppBar
    BarBackgroundColor="Black"
    BarBackgroundImage="background.gif" />
```



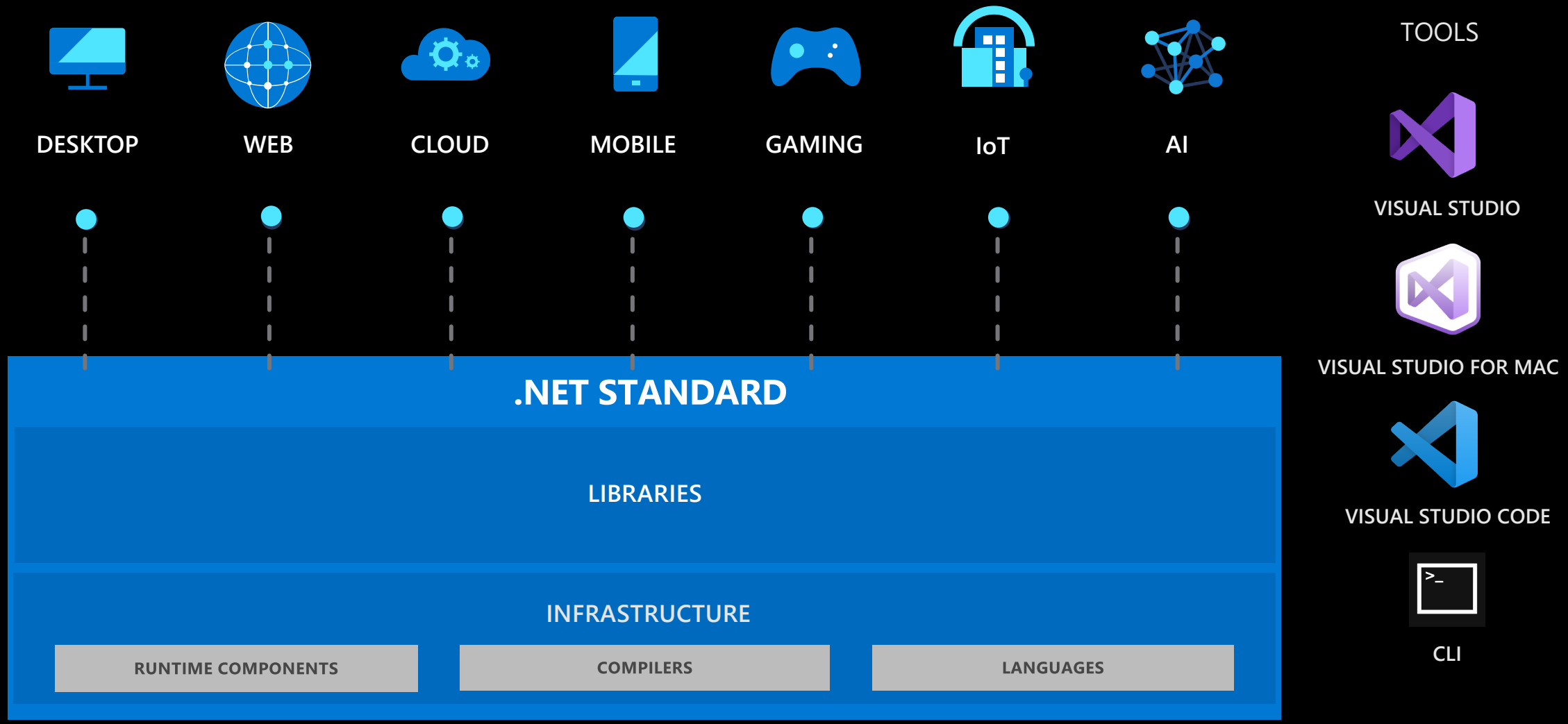
Release Schedule

aka.ms/xf-roadmap

Minor releases every 6 weeks
Service releases as needed

Xamarin.Forms 4.7 June 2020
Xamarin.Forms 4.8 August 2020
Xamarin.Forms 4.9 September 2020
Xamarin.Forms 5.0 November 2020
...and on and on

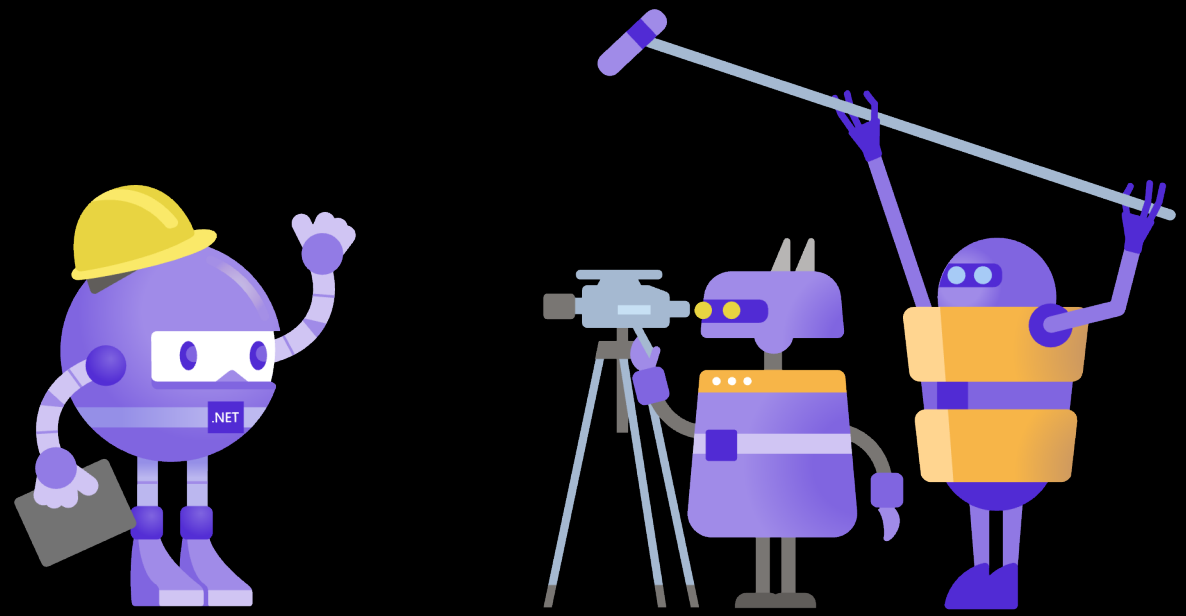
.NET – A unified platform





Time to Evolve

Journey to one .NET



One .NET vision – .NET 5 to 6 "wave"

.NET Framework

.NET

Mono / Xamarin

Single SDK, one BCL, unified toolchain

Cross-platform native UI

Cross-platform web UI

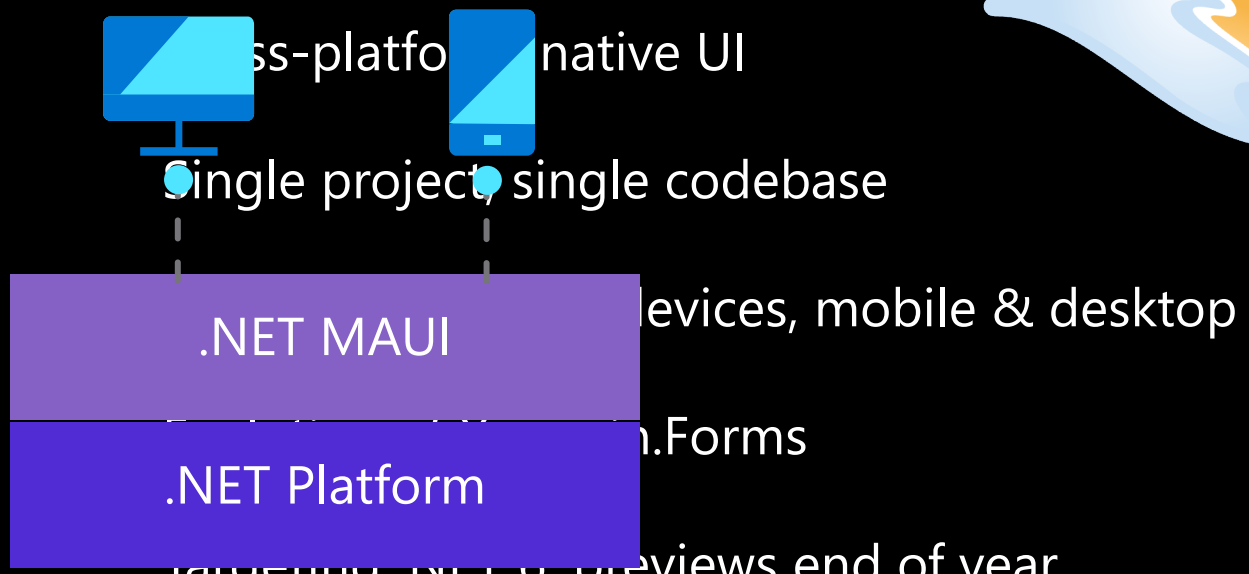
Cloud native investments

Continue improvements in speed, size, diagnostics, Azure services

.NET has the best of breed solutions for all modern workloads

Introducing .NET Multi-platform App UI





Build beautiful, native UI for any device

github.com/dotnet/maui

Microsoft

https://aka.ms/Build2020AppDev-DotNet

Maddy Leger and David Ortinau

Building UI with .NET MAUI

XAML and MVVM you know and love

```xaml
<StackLayout>
    <Label Text="Welcome to Maui!" />

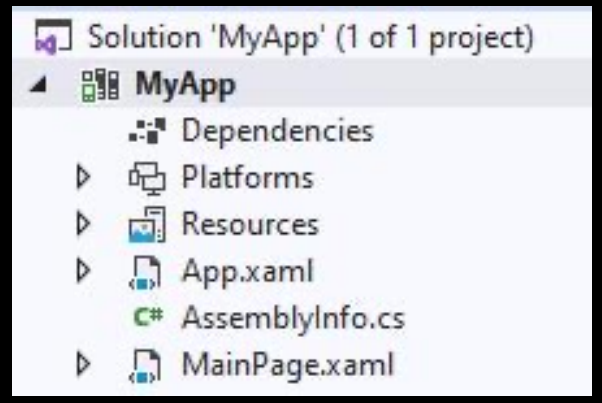
    <Button Text="{Binding Text}"
            Clicked="{Binding ClickCommand}" />
</StackLayout>
```

```
0 references
public ICommand ClickCommand { get; }
1 reference
public string Text { get; set; } = "Click me";

int count = 0;
0 references
void ExecuteClickCommand()
{
    count++;
    Text = $"You clicked {count} times.";
}
```

Vision Demo: .NET Multi-platform App UI

MAUI: Cross-platform development, simplified

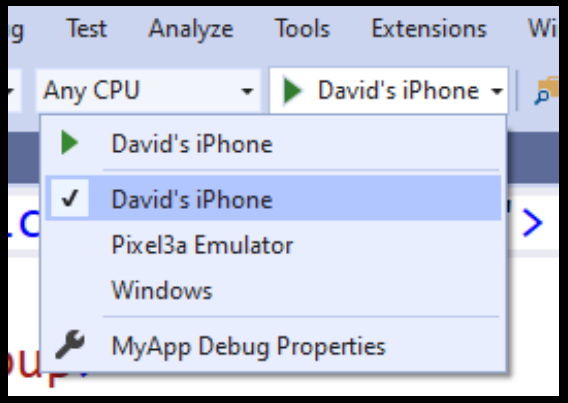


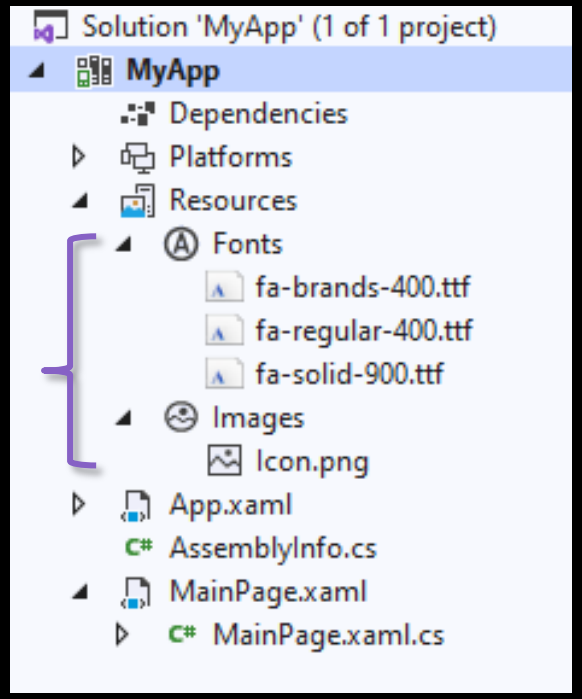

```
<Project Sdk="Microsoft.NET.Sdks">

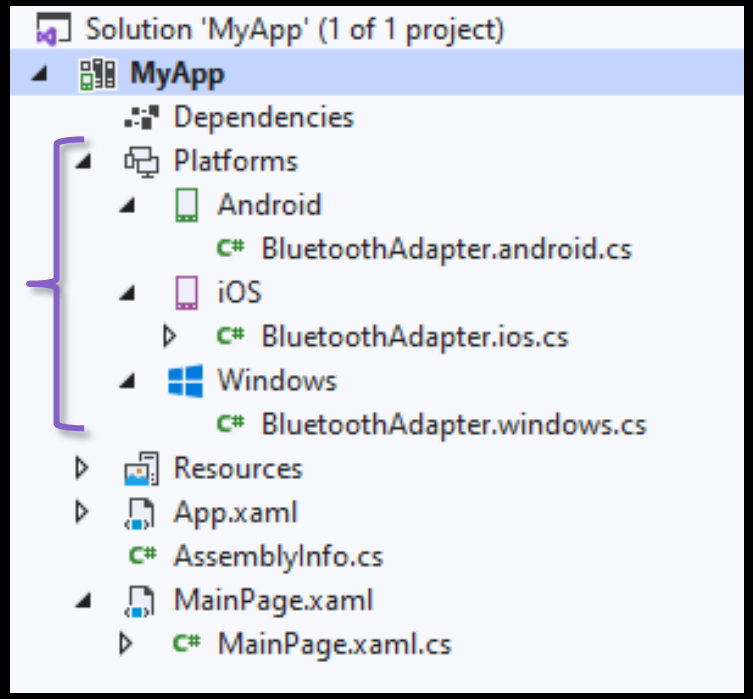
    <PropertyGroup>
        <TargetFrameworks>net5.0-ios;net5.0;net5.0-android</TargetFrameworks>
    </PropertyGroup>

    <ItemGroup>
        <PackageReference Include="System.Maui" />
        <PackageReference Include="System.Device" />
    </ItemGroup>

</Project>
```







.NET MAUI

System.Maui *evolution of Xamarin.Forms*
System.Devices *evolution of Xamarin.Essentials*

File | New
Multi-platform App UI (.NET)

CLI
```
dotnet new maui
```

.NET for iOS and Android

Xamarin.iOS --> .NET for iOS
Xamarin.Android --> .NET for Android

Namespaces & APIs unchanged from existing UIKit & Android APIs

File | New (SDK style)

CLI

```
dotnet new ios
dotnet new android
```

Building the Base

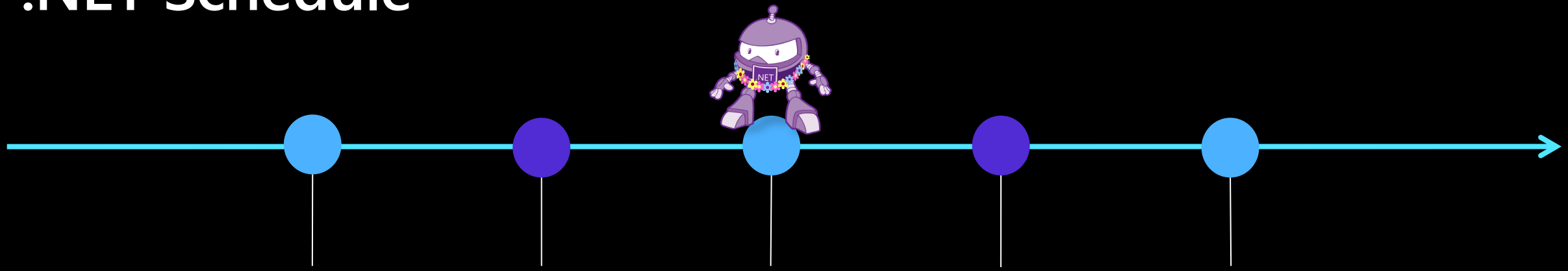
- Renderer re-architecting
  - New methodology
  - New organization
- Layout re-architecting
- Remove obsoletes
- Remove “slow” renderers
- Remove pre-AppCompat
- Drop Properties for Essentials
- Drop DependencyService for Microsoft.Extensions.DependencyInjection

- Drop .NET Standard pre-2.0
- Drop unsupported vendor API levels
- Cross-Platform Lifecycle
- Fix “AndExpand”
- Fix “MinHeightRequest” & friends
- Add GlideX

- Migration Tool with try-convert
  - Xamarin -> .NET 6
  - Xamarin.Forms -> .NET MAUI

	Xamarin.Forms 4	.NET MAUI
Platforms		
Android	API 19+	API 21+
iOS	9 – 13	10+
Linux (GTK#)	Community	Community
macOS	Community	Microsoft
Tizen	Samsung	Samsung
Windows	UWP Microsoft WPF Community	Windows
Features		
Renderers	Tightly coupled to BindableObject	Loosely coupled, no Core deps
App Models	MVVM / RxUI	MVVM / RxUI / MVU
Single Project	No	Yes
Multi-Targeting	No	Yes
Multi-Window	No	Yes
Pixel Drawn Controls	No	Yes
Misc		
.NET	X.iOS, X.Android, UWP, …	.NET 6 +
Project System	Franken-Proj	SDK Style
dotnet CLI	No	Yes
Tools		
Visual Studio 2019	Yes	Yes
Visual Studio 2019 for Mac	Yes	Yes
Visual Studio Code	No	Yes

.NET Schedule



- .NET 5.0 release in November 2020
- .NET 6.0 Maui + Xamarin SDK support

ANNOUNCING

.NET Conf 2020

November 10 – 12

.NET 5 launch

Free, three-day, virtual event

Save the date!

www.dotnetconf.net



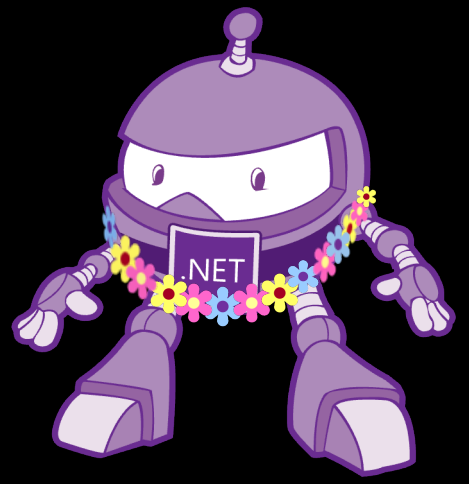
We want to hear from you!

Give us your feedback

aka.ms/maui_feedback

Discuss Maui proposals

github.com/dotnet/maui



Thanks for joining!

Give us your feedback

aka.ms/dotnet5feedback

